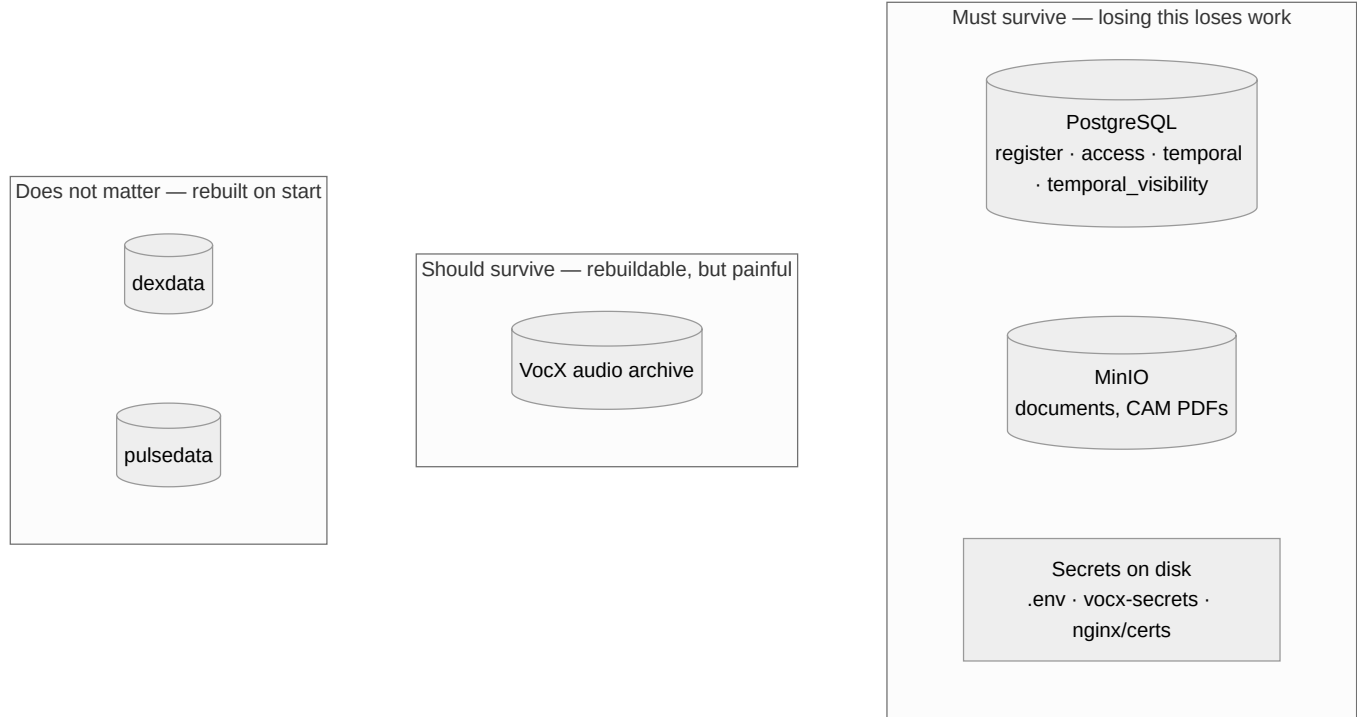


09 — Backup & Restore

Audience: whoever is accountable for not losing the book. **Companion docs:** [10 Upgrade & rollback](#) · [02 Deployment](#) · [13 Operations](#)

1. What has to survive



Asset	Where	Backed up by	Losing it means
PostgreSQL	<code>pgdata</code> volume	<code>pgbackup</code> nightly and <code>prism-deploy.sh backup</code>	Total data loss
MinIO objects	<code>miniodata</code> volume	volume snapshot only — no automatic job	Documents gone; rows still reference them
<code>deploy/compose/.env</code>	bind mount	<code>prism-deploy.sh</code> secrets snapshot	Stack misconfigures; every secret lost
<code>deploy/vocx-secrets/</code>	bind mount	secrets snapshot	Every user must re-authorise Google
<code>deploy/nginx/certs/</code>	bind mount	secrets snapshot	The edge will not start
VocX audio archive	volume	volume snapshot only	Recordings unrecoverable; transcripts survive in the Register
<code>dexdata</code> , <code>pulsedata</code>	volumes	—	Rebuilt on next start

The honest gap: MinIO objects and the VocX audio archive have **no scheduled backup** today. The database does. If document bytes matter to you — and for a lender they do — add a volume snapshot or an `mc mirror` to off-box storage. \$7 gives both.

2. The three secret paths — say them out loud

```
deploy/compose/.env
deploy/vocx-secrets/
deploy/nginx/certs/
```

These are **never in git and never in a delivery archive**. They live only on the VM. That makes the secrets snapshot the *only* copy outside the running tree, and it makes verifying any zip you produce mandatory:

```
unzip -l "$ZIP" | grep -Ei "client_secret|token\.json|vocx-
secrets/*.json|vocx_tokens|compose/.env"
# exit code 1 (no matches) = clean. Any output at all = do not send that file.
```

3. The automatic database backup

`pgbackup` runs under the `backup` compose profile:

```
pgbackup:
  image: postgres:16-alpine
  profiles: ["backup"]
  environment:
    KEEP_DAYS: "${PGBACKUP_KEEP:-14}"
  volumes:
    - pgbackups:/backups
  command:
    - |
      while true; do
        ts=$(date +%Y%m%d-%H%M%S)
        pg_dumpall --clean --if-exists | gzip > "/backups/prism-$ts.sql.gz"
        find /backups -name 'prism-*.sql.gz' -mtime "+$KEEP_DAYS" -delete
        sleep 86400
      done
```

Property	Value
Frequency	Every 24 h, counted from container start
Scope	<code>pg_dumpall</code> — all four databases , roles included
Retention	14 days (<code>PGBACKUP_KEEP</code>)
Location	The <code>pgbackups</code> volume, on the same host

Three things to be clear-eyed about:

1. **It is not a cron.** It is a `sleep 86400` loop, so a container restart moves the backup time. Do not expect it at a fixed hour.

2. **It stays on the same machine.** A host loss loses the backups with the data. Copy them off-box (§7).
3. **Nothing verifies restorability.** A dump that exists is not a dump that restores. Do the drill in §6.

```
# Confirm it is running and see what it has
docker compose -f deploy/compose/docker-compose.yml --profile backup ps pgbackup
docker compose -f deploy/compose/docker-compose.yml run --rm -v compose_pgbackups:/b \
  postgres:16-alpine ls -lh /b
```

4. The deploy-script backup (the one taken before every upgrade)

```
./prism-deploy.sh backup      # a dump + a secrets snapshot, and nothing else
```

This runs automatically as the **first step** of every `upgrade`. Rule 1 of the script: *"NOTHING IS TOUCHED UNTIL A BACKUP EXISTS."*

The database dump — and how it refuses to lie to you

```
dc "$LIVE" exec -T postgres pg_dumpall -U prism | gzip > "$dump"
gzip -t "$dump"                                || die "the dump is corrupt (gzip -t failed)"
(( bytes > 100000 ))                            || die "the dump is only ${bytes} bytes – refusing to treat
that as a backup"
```

Three checks, each closing a real failure: postgres not running at all, a truncated gzip stream, and a dump that "succeeded" but is empty. **A backup that is not verified is not a backup**, and this is what that principle looks like in code.

The secrets snapshot

```
SECRET_PATHS=(deploy/compose/.env deploy/vocx-secrets deploy/nginx/certs)
tar -czf "$tarball" -C "$LIVE" "${present[@]}"
chmod 600 "$tarball"
```

If the tar fails — most commonly a root-owned `nginx/certs/tls.key` while running as `ubuntu` — the script stops and names the file and both remedies. It does **not** proceed with a partial snapshot.

Where backups land

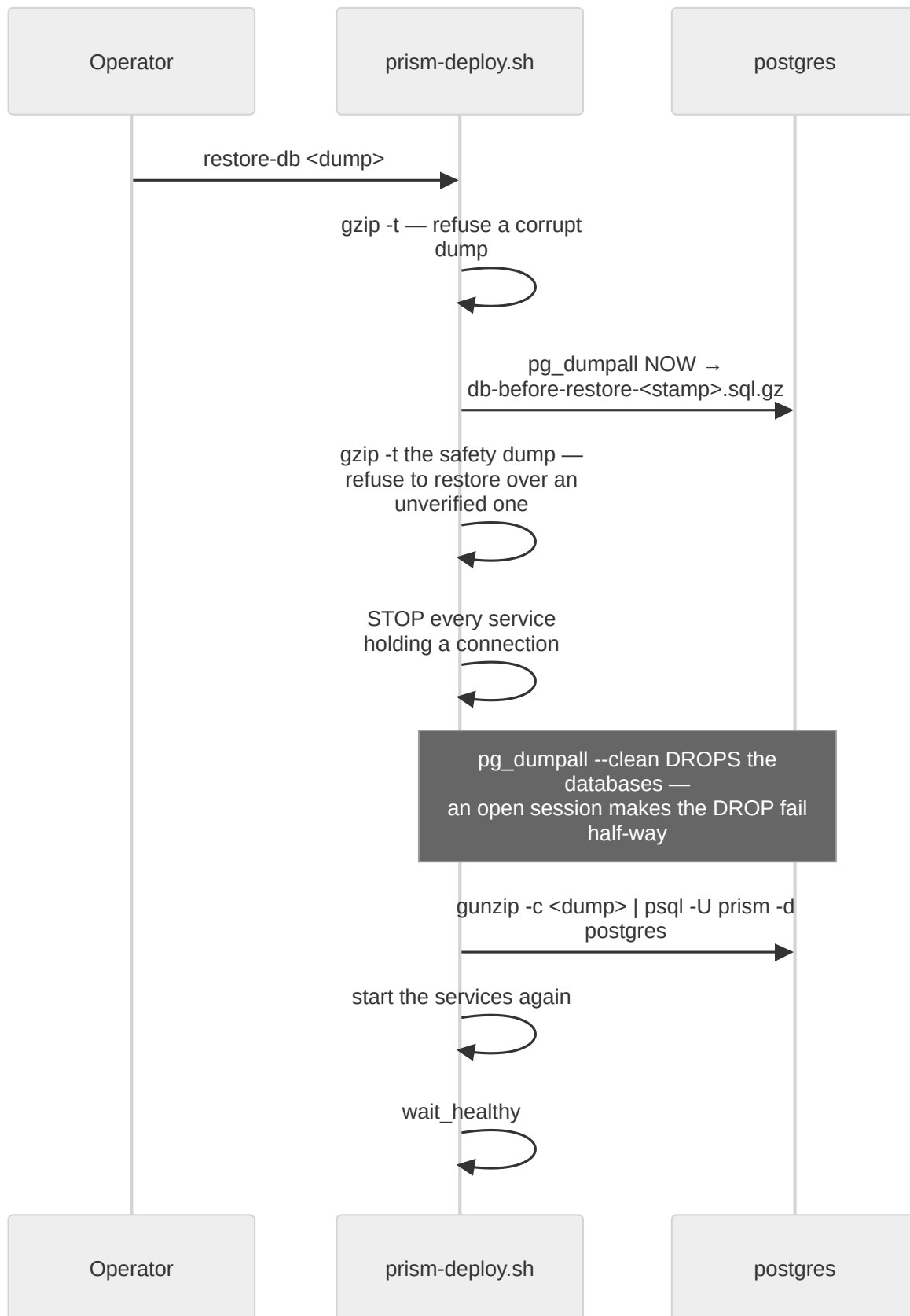
```
$PRISM_ROOT/backups/
├─ db-20260819-041500.sql.gz          ← pre-upgrade dump
├─ secrets-20260819-041500.tar.gz     ← chmod 600
├─ images-20260819-041500.tsv        ← the rollback image map
├─ db-before-restore-<stamp>.sql.gz   ← safety dump taken before any restore
└─ deploy-20260819-041500.log
```

Retention: `PRISM_KEEP_BACKUPS` (default 10).

5. Restoring

The full-database restore

```
./prism-deploy.sh restore-db ~/aug_11/backups/db-20260819-041500.sql.gz
```



Two design points worth keeping:

- **A restore replaces, so the present is captured first.** The step is itself reversible.
- **Connection holders are stopped explicitly** — `register access gateway atlas vocx pulse workflows notifier orchestrator temporal temporal-ui`. A half-applied `--clean` restore is a far worse state than a refused one.

Restoring by hand

```
cd ~/aug_11/prism
docker compose -f deploy/compose/docker-compose.yml stop \
  register access gateway atlas vocx pulse workflows notifier orchestrator temporal

gunzip -c /path/to/prism-20260819.sql.gz | \
  docker compose -f deploy/compose/docker-compose.yml exec -T postgres psql -U prism -d postgres

docker compose -f deploy/compose/docker-compose.yml start \
  register access gateway atlas vocx pulse workflows notifier orchestrator temporal
docker compose -f deploy/compose/docker-compose.yml exec nginx nginx -s reload
```

That last line matters — see [13 Operations](#) on stale upstream IPs.

Restoring the secrets

```
tar -xzf ~/aug_11/backups/secrets-20260819-041500.tar.gz -C ~/aug_11/prism
ls -l ~/aug_11/prism/deploy/compose/.env \
  ~/aug_11/prism/deploy/vocx-secrets \
  ~/aug_11/prism/deploy/nginx/certs
```

`upgrade` does this automatically into the new tree, and **hard-fails** if `.env` is absent afterwards.

6. The restore drill — do this quarterly

An untested backup is a hope. The drill takes about twenty minutes and is the only thing that converts it into a recovery position.

```
# 1. On a scratch machine (or a second compose project name), bring up postgres alone
docker run -d --name drill -e POSTGRES_USER=prism -e POSTGRES_PASSWORD=x postgres:16-alpine

# 2. Restore the newest dump
gunzip -c prism-20260819-041500.sql.gz | docker exec -i drill psql -U prism -d postgres

# 3. Prove the book is there
docker exec -i drill psql -U prism -d register -c \
  "select count(*) from entities; select count(*) from deals;
  select count(*) from lending_tracker; select count(*) from interactions;"

# 4. Prove identity came too
docker exec -i drill psql -U prism -d access -c \
  "select count(*) from users; select count(*) from access_grants;"

# 5. Tear down
docker rm -f drill
```

Record the row counts. A dump whose counts have quietly collapsed is the failure you are drilling for.

7. Getting backups off the box (recommended, not yet built)

Everything above keeps backups on the same host. Two additions close that:

```
# Database dumps → S3, nightly
aws s3 sync ~/aug_11/backups/ s3://evam-prism-backups/${hostname}/ \
  --exclude '*' --include 'db-*.sql.gz' --include 'secrets-*.tar.gz'

# MinIO objects → S3 (or another MinIO), nightly
docker compose -f deploy/compose/docker-compose.yml exec -T minio \
  mc mirror --overwrite local/prism s3remote/evam-prism-documents
```

If you add the secrets tarball to an off-box location, that location now holds every secret PRISM has. Encrypt it and restrict who can read it — this is not a place to be casual.

8. RPO and RTO, honestly stated

	Today	With off-box sync
RPO (data you can lose)	Up to 24 h — the gap between nightly dumps. A pre-upgrade dump narrows it only around upgrades.	Same 24 h, but survives host loss
RTO (time to recover)	~15–30 min from a local dump: restore, start, verify	Add transfer time
Host loss	Total loss — backups are on the same volume	Recoverable
Document bytes	No backup — MinIO is unprotected	Recoverable

If a 24-hour RPO is not acceptable for the book, the honest fix is WAL archiving (`archive_mode=on` plus `pg_receivewal` or a base-backup tool) rather than more frequent `pg_dumpall`. That is a real change and it is not in place today.

9. What is *not* backed up, and why it is usually fine

Not backed up	Why acceptable	When it is not
Temporal history	It rides inside <code>pgdata</code> , so <code>pg_dumpall</code> does capture it	—
<code>dexdata</code>	Dex re-seeds on start; identities live upstream	If Dex is your only user store
<code>pulsedata</code>	Rebuilt by the next scan	Never
Container images	Rebuildable from source; <code>prism-deploy.sh</code> also tags them for rollback	If the source archive is lost too
VocX audio	Transcripts and structured data are in the Register	If recordings are evidence you must retain

That last row is a policy question, not a technical one. If audio is retained evidence, back the volume up.

10. Checklist

Weekly

- ☐ `docker compose --profile backup ps pgbackup` — still running?
- ☐ Newest dump is < 24 h old and > 100 KB
- ☐ Disk headroom on the backup volume

Quarterly

- ☐ Run the restore drill (§6) and record the row counts
- ☐ Confirm the secrets tarball opens and contains all three paths
- ☐ Confirm the TLS certificate in the snapshot is the current one

Before anything risky

- ☐ `./prism-deploy.sh backup`
- ☐ Note the dump path — `rollback --with-db` and `restore-db` both need it